

# Hospital Appointment & Patient Care Management System

Problem Statement & Architectural Approach

DBMS Project Milestone — Review 1

Database Platform: PostgreSQL 16 on Neon Cloud

# The Problem: Structural Breakdowns in Healthcare Operations

- • Schedule Collisions & Doctor Double-Booking:
  - - Front desks frequently book overlapping patients into the same consultation slot, leading to long waiting queues and doctor burnout.
- • Fragmented & Inaccessible Patient Records:
  - - Clinical history, past diagnoses, lab reports, and medications are stored across disconnected paper slips, preventing a unified longitudinal chart at point-of-care.
- • Inpatient Bed Allocation Bottlenecks:
  - - Lack of real-time bed state tracking results in double-allocation of ward beds, admission delays, and unrecorded vacancies.
- • Uncoordinated Clinical Orders & Revenue Leakage:
  - - Lab tests and prescriptions are unlinked from billing ledgers, causing untracked dispensing and severe financial leakage.

# Why Existing Systems Fail: Root Cause Analysis

- • Absence of Database-Level Concurrency & Integrity Enforcement:
  - - Legacy systems rely on manual checks or fragile client-side scripts rather than database ACID constraints.
- • Unnormalized Data Architecture (Anomalies & Redundancy):
  - - Multi-valued fields (e.g., comma-separated drug lists) violate 1NF, while transitive dependencies create update and deletion anomalies.
- • Disconnected Departmental Silos:
  - - Front-desk scheduling, clinical consultations, pathology labs, and billing operate on isolated tables or paper records.
- • Vulnerability to Invalid Data:
  - - Negative invoices, overlapping bed occupancies, and invalid physiological bounds are accepted without engine-level rejection.

# Our Approach: Unified Relational Architecture (PostgreSQL & 3NF)

- • Centralized Relational Schema (16 Normalized Entities in 6 Clusters):
  - - Provider Roster | Patient Appointments | Clinical Care & EMR | Diagnostics | Inpatient Wards | Financials & Billing
- • Strict Third Normal Form (3NF) Normalization:
  - - 1NF: Atomic line items (prescription\_item) eliminate multi-valued repeating groups.
  - - 2NF & 3NF: Single surrogate keys and isolated lookup tables eliminate partial and transitive functional dependencies.
- • Engine-Level Defensive Constraints (Active Database Protection):
  - - UNIQUE(doctor\_id, date, time) mathematically prevents appointment double-booking.
  - - Real-time Bed State Machine (AVAILABLE, OCCUPIED, MAINTENANCE) prevents double-occupancy.
  - - Domain CHECK bounds enforce positive balances, chronological stay validity, and valid patient vitals.

# Strategic Outcomes: Measurable Operational Impact

- • Zero-Collision Scheduling:
  - - 100% elimination of double-booked consultation slots via engine-enforced uniqueness.
- • Point-of-Care Longitudinal EMR:
  - - Clinicians access unified consultation notes, ICD-10 diagnoses, prescribed drugs, and lab results via indexed multi-table joins.
- • Real-Time Inpatient Bed Visibility:
  - - Instant bed occupancy percentages across ICU, Semi-Private, and General wards with exclusive occupancy guarantees.
- • Closed-Loop Financial Reconciliation:
  - - Itemized bills aggregate consultation, lab investigations, and bed per-diem charges into a single ledger with complete multi-tender audit trails.
- • Review 1 Foundation:
  - - Complete conceptual ERD, production PostgreSQL DDL, and realistic seed data ready for Review 2 (Triggers & Views).